KAIZEN MATHS

Lesson starter

Function Notation, Composite and Inverse Functions

Inverse Functions

Name _____

S1

Find the inverse of f(x) = 2x + 3.

S2

For f(x) = -4x - 1, find $f^{-1}(-29)$.

S3

Find the inverse of f(x) = 3x + 2.

S4

For f(x) = -3x + 4, find $f^{-1}(7)$.

Show your working

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Exit ticket

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E2

For f(x) = -3x + 3, find $f^{-1}(-12)$.

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ALG-2F88-L3 · E

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Teacher answers & working

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S1

Find the inverse of f(x) = 2x + 3.

Answer:

$(x - 3)/2$

Working:

Step 1: Write the function as an equation using y.

$y = 2x + 3.$

Step 2: Swap x and y because the inverse reverses the input and output.

$x = 2y + 3.$

Step 3: Solve this equation for y.

$y = (x - 3)/2.$

Step 4: Replace y with inverse-function notation.

$f^{-1}(x) = (x - 3)/2.$

S2

For f(x) = -4x - 1, find $f^{-1}(-29)$.

Answer:

7

Working:

Step 1: $f^{-1}(-29)$ asks for the input that gives the output -29.

Step 2: Set f(x) equal to -29.

$-4x - 1 = -29.$

Step 3: Use inverse operations to solve for x.

$-4x = -28$

$x = 7.$

Step 4: Therefore $f^{-1}(-29) = 7$.

Teacher answers & working

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S3

Find the inverse of f(x) = 3x + 2.

Answer:

$(x - 2)/3$

Working:

Step 1: Write the function as an equation using y.

$y = 3x + 2.$

Step 2: Swap x and y because the inverse reverses the input and output.

$x = 3y + 2.$

Step 3: Solve this equation for y.

$y = (x - 2)/3.$

Step 4: Replace y with inverse-function notation.

$f^{-1}(x) = (x - 2)/3.$

S4

For f(x) = -3x + 4, find $f^{-1}(7)$.

Answer:

-1

Working:

Step 1: $f^{-1}(7)$ asks for the input that gives the output 7.

Step 2: Set f(x) equal to 7.

$-3x + 4 = 7.$

Step 3: Use inverse operations to solve for x.

$-3x = 3$

$x = -1.$

Step 4: Therefore $f^{-1}(7) = -1$.

Teacher answers & working

Function Notation, Composite and Inverse Functions · Inverse Functions

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E1

Find the inverse of f(x) = -5x.

Answer:

$-x/5$

Working:

Step 1: Write the function as an equation using y.

$y = -5x$.

Step 2: Swap x and y because the inverse reverses the input and output.

$x = -5y$.

Step 3: Solve this equation for y.

$y = -x/5$.

Step 4: Replace y with inverse-function notation.

$f^{-1}(x) = -x/5$.

E2

For f(x) = -3x + 3, find $f^{-1}(-12)$.

Answer:

5

Working:

Step 1: $f^{-1}(-12)$ asks for the input that gives the output -12.

Step 2: Set f(x) equal to -12.

$-3x + 3 = -12$.

Step 3: Use inverse operations to solve for x.

$-3x = -15$

$x = 5$.

Step 4: Therefore $f^{-1}(-12) = 5$.

Teacher answers & working

Function Notation, Composite and Inverse Functions · Inverse Functions

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E3

Find the inverse of f(x) = 2x - 7.

Answer:

$(x + 7)/2$

Working:

Step 1: Write the function as an equation using y.

$y = 2x - 7.$

Step 2: Swap x and y because the inverse reverses the input and output.

$x = 2y - 7.$

Step 3: Solve this equation for y.

$y = (x + 7)/2.$

Step 4: Replace y with inverse-function notation.

$f^{-1}(x) = (x + 7)/2.$